Calculus 12 - Limits and Early Derivatives



— SINE TIMORE AUT FAVORE —

ST. GEORGE'S SCHOOL

Name : _____

Teacher's name: _____

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Last Updated: October 23, 2026

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1 Asymptotes

The concept of asymptotes is more or less identical to Rational Functions in Pre-Calculus 12. In Calculus 12, we won't be limited to polynomial functions. Understanding the asymptotic behaviour of a curve can tell us a lot about how it looks. We will break down these sections into vertical asymptotes, horizontal asymptotes, and later on, slant asymptotes.

Vertical Asymptotes

For the functions in the form of

$$y = \frac{f(x)}{g(x)}$$

We can locate either vertical asymptotes or points of discontinuity (holes) by the following

- $x = a$ is a vertical asymptote if $g(a) = 0$ and $f(a) \neq 0$
- $x = a$ is a point of discontinuity if $g(a) = 0$ and $f(a) = 0$

Example 1.1. Identify any vertical asymptotes or points of discontinuities for the following curve

$$y = \frac{x^2 - 9}{(x+3)\sqrt{x+8}}$$

Find the values that make the denominator and numerator zero:

Identify which value is a point of discontinuity and which is a vertical asymptote:

Simplify y to find the coordinate of the point of discontinuity:

Concluding Statements:

One-Sided Limits

Once a vertical asymptote is established, we can look at the vertical asymptote from each side. Because it is a vertical asymptote, at least one of the sides (left or right) must approach positive or negative infinity.

Example 1.2. Complete the table of values for the following curve

$$y = \frac{1}{x}$$

x	0.1	0.01	0.001	-0.1	-0.01	-0.001
y						

We can conclude that the $\lim_{x\to 0^-} \frac{1}{x}$ approaches $\square$ and $\lim_{x\to 0^+} \frac{1}{x}$ approaches $\square$

We can use the notation $\lim_{x\to 0^-} \frac{1}{x} = -\infty$ and $\lim_{x\to 0^+} \frac{1}{x} = \infty$ with the understanding that this is describing behaviour, and that infinity is **not** a number, and thus, should not be used to evaluate algebraically.

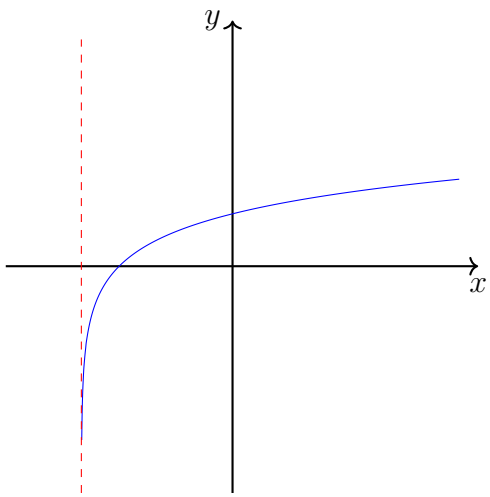
Example 1.3. Find the one-sided behaviour of the vertical asymptote in Example 1.1.

$$y = \frac{x^2 - 9}{(x+3)\sqrt{x+8}}$$

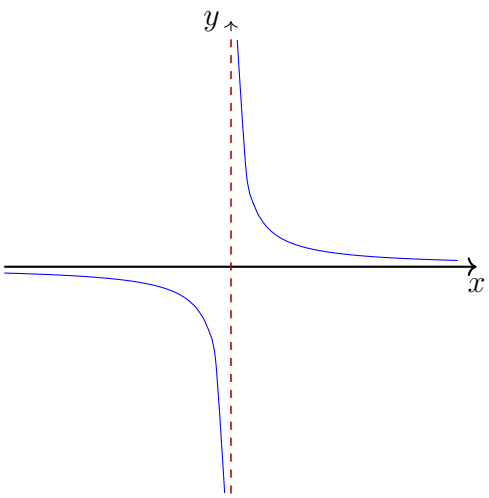
Graphical Examples - Vertical Asymptotes

Below are some different types of vertical asymptotes.

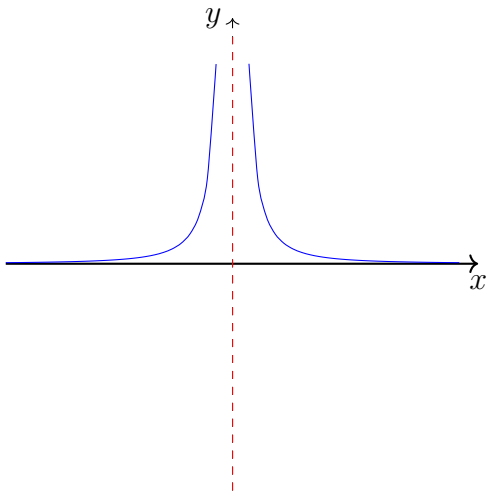
a)



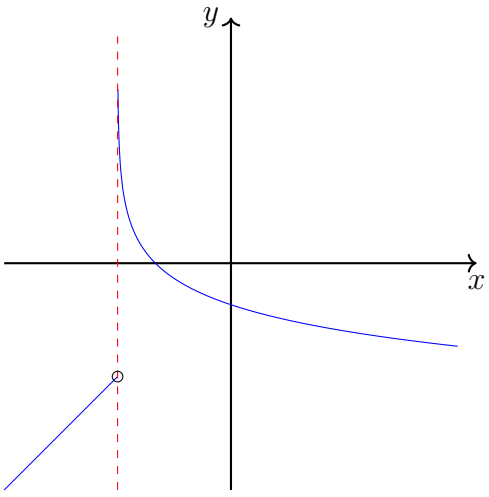
b)



c)



d)



Horizontal Asymptotes

Horizontal asymptotes were likely first introduced when you studied the reciprocal function, $y = \frac{1}{x}$, or when you studied rational functions, $y = \frac{f(x)}{g(x)}$.

For rational functions (numerator and denominator are polynomials), we can simply compare the degrees of the numerator and denominator. Assume we have a rational function of the form

$$y = \frac{f(x)}{g(x)}$$

- If $deg(f) > deg(g)$, there will be no horizontal asymptote.
- If $deg(f) < deg(g)$, there will be a horizontal asymptote at $y = 0$.
- If $deg(f) = deg(g)$, there will be a horizontal asymptote at the fraction of their leading coefficients.

The formal definition of a horizontal asymptote is as follows.

If there are horizontal asymptotes for $f(x)$, then

$$\lim_{x \to \infty} f(x) = L$$

or

$$\lim_{x \to -\infty} f(x) = M$$

Where L, M are real constants

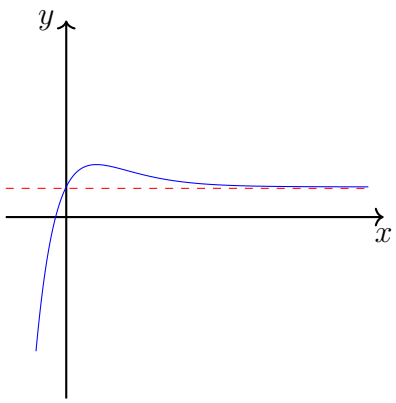
Example 1.4. Using an algebraic method, evaluate the following

$$\lim_{x \to \infty} \frac{4x^2 - x + 2}{6x^2 + 5x + 1}$$

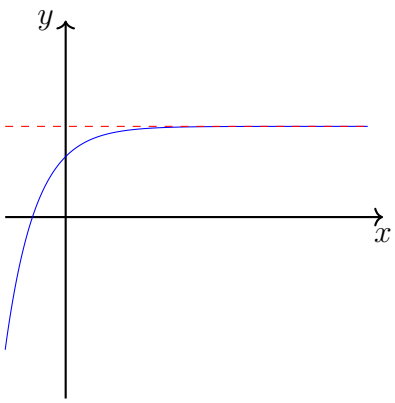
Graphical Examples - Horizontal Asymptotes

Below are some different types of horizontal asymptotes.

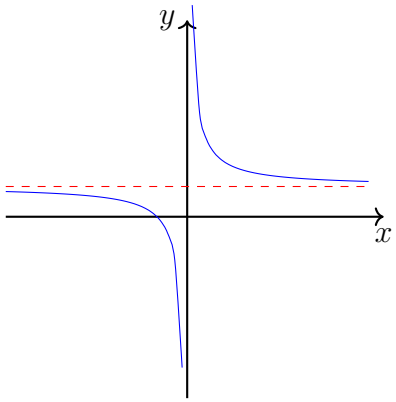
a)



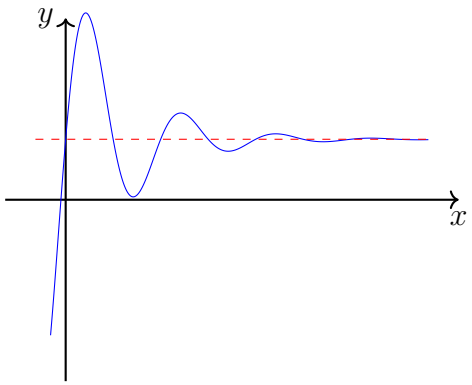
b)



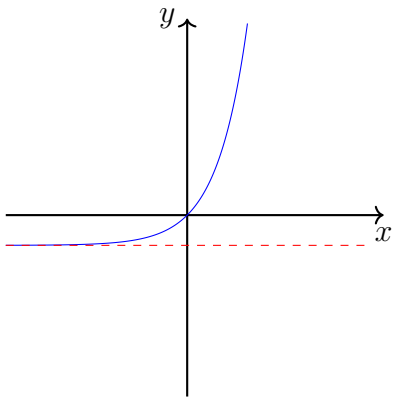
c)



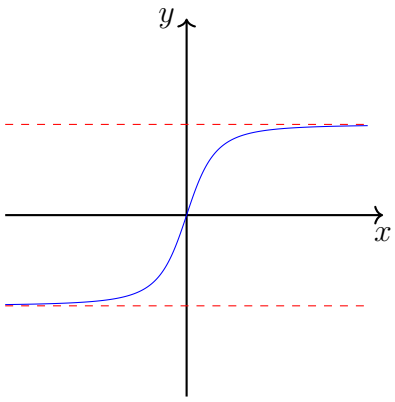
d)



e)



f)



Practice Exercises

1. Find the vertical asymptotes.

   (a) $y = \dfrac{2}{x+1}$

   (b) $y = \dfrac{1}{x^2-1}$

   (c) $y = \dfrac{2x}{3x-1}$

   (d) $y = \dfrac{x^3}{4x^2-9}$

   (e) $y = \dfrac{x^2-4}{x^2+x-6}$

   (f) $y = \dfrac{2x^2+x-3}{x^2-1}$

2. Find the horizontal asymptotes.

   (a) $y = \dfrac{x}{x+1}$

   (b) $y = \dfrac{4x+5}{3-2x}$

   (c) $y = \dfrac{1}{x^2-1}$

   (d) $y = \dfrac{x^2}{(x+2)^2}$

   (e) $y = 1 - \dfrac{x}{x^2+1}$

   (f) $y = \dfrac{16-2x^3}{9x^3-1}$

2 Concavity and Points of Inflection

Concavity can be considered how a function curves. Concavity can be found by analyzing the **second derivative**.

If $f''(x) > 0$, for all x in any interval I, then the function $f(x)$ is **concave up** on the interval I
If $f''(x) < 0$, for all x in any interval I, then the function $f(x)$ is **concave down** on the interval I

How the shape looks:

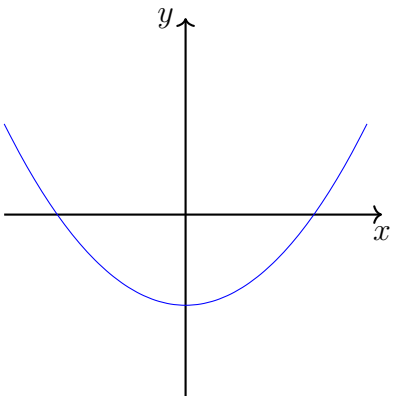


Figure 1: Concave Up

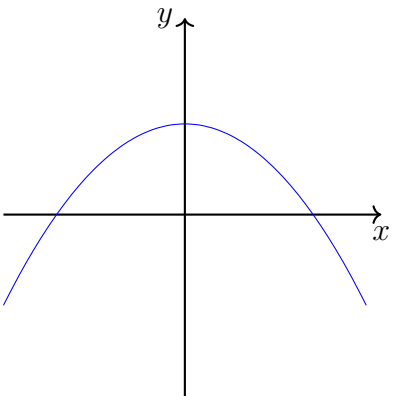
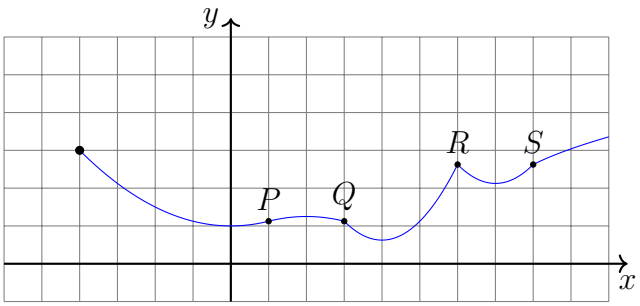


Figure 2: Concave Down

We define a **point of inflection** or an **inflection point** as a point where the concavity changes. Inflection points can be found when $f''(x) = 0$ or when $f''(x)$ does not exist for x values in the domain of $f(x)$. If the second derivative is zero or undefined, but the concavity does not change over the point, then it is not an inflection point. This is very similar to the First Derivative Test. A number line helps.

Example 2.1. Given the graph below. Identify the intervals of concave up, concave down, and the points of inflection.



Example 2.2. For $y = x^3 - 3x^2 + 4x - 5$, find

(a) The interval(s) of concave up

(b) The interval(s) of concave down

(c) Any inflection points

Example 2.3. For $y = \dfrac{x}{x^2 + 1}$, find

(a) The interval(s) of concave up

(b) The interval(s) of concave down

(c) Any inflection points

Practice Exercises

1. Find the intervals on which the curve is concave upward or concave downward, and state the points of inflection.

   (a) $y = 6x^2 - 12x + 1$

   (b) $y = x^4 - 2x^3 + x - 2$

   (c) $y = \dfrac{1}{x-1}$

   (d) $y = \dfrac{1-x^2}{x^3}$

2. For each of the following functions,
   (a) Find the intervals of increase or decrease,
   (b) find the local maximum and minimum values,
   (c) find the intervals of concavity
   (d) find the points of inflection

   i. $y = x\sqrt{x^2 + 4}$

   ii. $y = 3x^{\frac{2}{3}} - 2x$

3 The Second Derivative Test

The second derivative test is another way to test maximum or minimum points. It is relatively simple to use if you have the second derivative handy, however, the first derivative test can provide more information.

Recall the shape of a concave up and a concave down graph.

The Second Derivative Test

If $f'(a) = 0$ and $f''(a) > 0$, then $x = a$ is a local minimum.
If $f'(a) = 0$ and $f''(a) < 0$, then $x = a$ is a local maximum.
If $f'(a) = 0$ and $f''(a) = 0$, then SDT is inconclusive.

Example 3.1. Find the local maximum and minimum values of $f(x) = x^3 - 12x + 5$

Practice Exercises

1. Use the second derivative test to find the local maximum and minimum values of each function

   (a) $f(x) = 1 + 3x^2 - 2x^3$

   (b) $g(x) = x^4 - 6x^2 + 10$

   (c) $h(x) = \dfrac{x}{x^2 + 9}$

   (d) $i(x) = x + 3x^{\frac{2}{3}}$

4 Slant Asymptotes

Occasionally, the asymptotic behaviour of a curve is a slant. If a curve of the form $y = \dfrac{f(x)}{g(x)}$ has a slant asymptote, then the **quotient** of $f(x) \div g(x)$ is a line.

If $f(x)$ and $g(x)$ are polynomials, then that would mean $f(x)$ is one degree higher than $g(x)$. Below is the graph of $y = \dfrac{x^2 - 8}{3x + 1}$

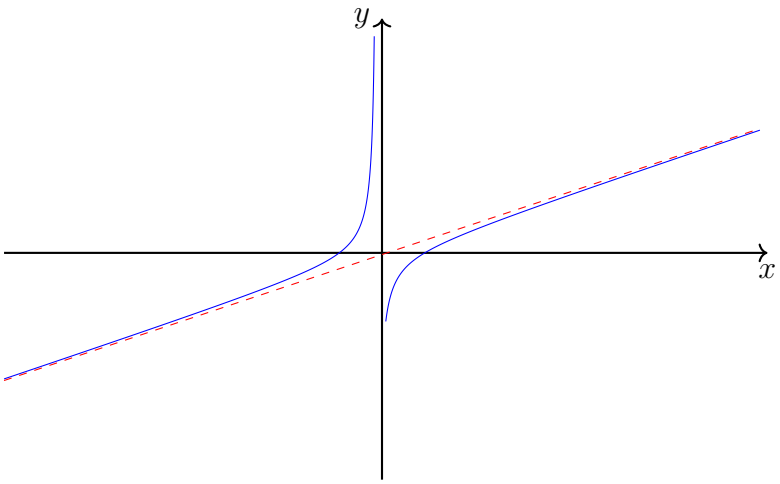


Figure 3: Slant Asymptote

We can see that the curve does not have a horizontal asymptote, but as the function approaches infinity (either side), it looks more like a straight line than a curve. We can find the equation of the slant asymptote by doing long division.

Note: It is not possible for a function to have a horizontal asymptote as well as a slant asymptote.

Example 4.1. Find the slant asymptote for the following function

$$f(x) = \frac{2x^3 - 3x^2 + x - 3}{x^2 + 1}$$

Practice Exercises

1. Find the equation of the slant asymptote.

(a) $y = \dfrac{2x - x^2 - 1}{x}$

(b) $y = \dfrac{x^3 - 1}{x^2}$

(c) $y = \dfrac{3x^2 + 4x + 2}{x + 1}$

(d) $y = \dfrac{4x^2}{2x + 1}$

(e) $y = \dfrac{x^3 + 4x^2 + 5x + 16}{x^2 + 4}$

(f) $y = \dfrac{x + x^2 - x^4}{x^3 - 1}$

5 Curve Sketching

This section is where we put everything together. We will be using all the skills learned so far to do quite an accurate sketch without the use of graphing technology. Let's summarize what each of the following can tell you.

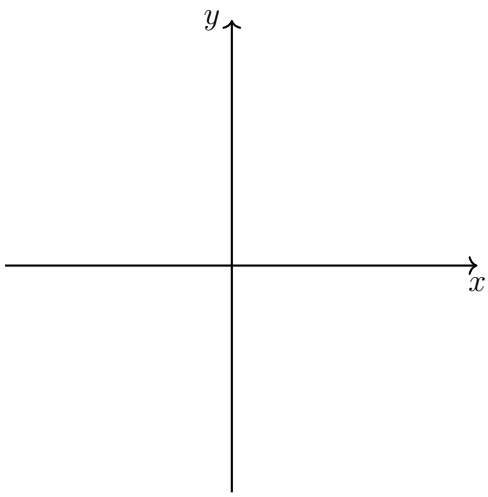
Function	Information Received
$y = f(x)$	- Domain - Any asymptotes (vertical/horizontal/slant) or holes - Intercept Points - y values of any max/min or inflection points
$y = f'(x)$	- Critical numbers (zero slopes and undefined) - Intervals of increase and decrease
$y = f''(x)$	- Inflection points (concavity changes) - Intervals of concave up and down

Curve sketching problems can take a lot of care and attention, and the process is quite long. Here are some suggest steps to go through.

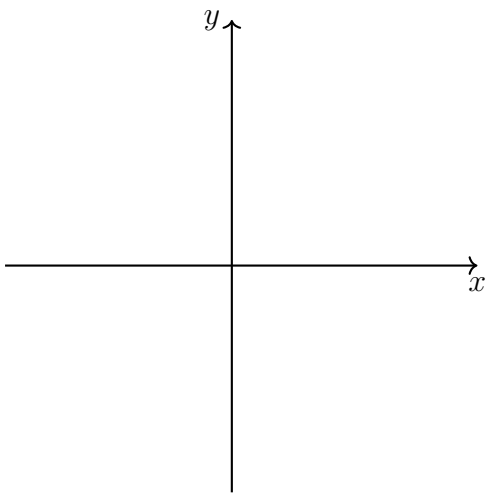
1. Identify the domain
2. Find any intercepts, asymptotes, or points of discontinuity
3. Find the first and second derivatives
4. Find critical numbers of the first and second derivatives
5. Plot two number lines, one for f' and one for f''
6. Put all critical numbers into each and vertically divide them into parts. Consider the domain here.
7. Find y-values of any identified max/min, inflection.
8. Make a general shape of each interval based on concavity and increase/decrease
9. Start graphing! (Asymptotes, special points, sketch left to right connecting all key points)

To do this successfully, you must get the **first and second derivative** correct!

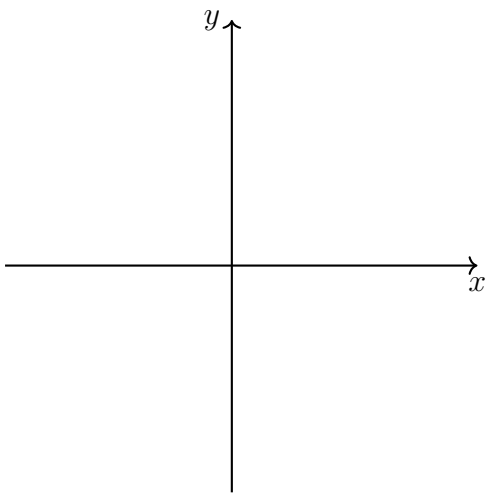
Example 5.1. Follow the procedures to sketch $y = 3x^5 - 5x^3$



Example 5.2. Follow the procedures to sketch $y = \dfrac{x^2}{1-x^2}$



Example 5.3. Follow the procedures to sketch $y = x\sqrt{2-x}$

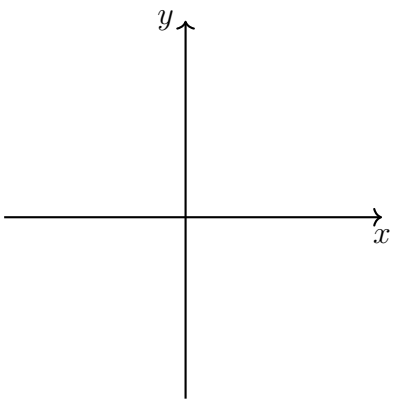


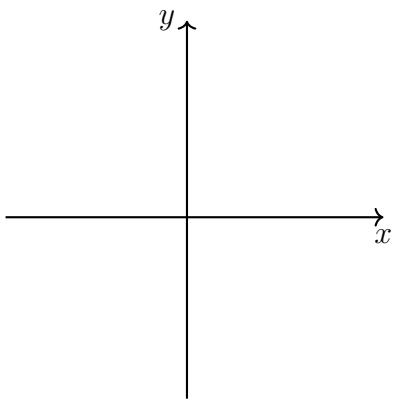
Practice Exercises

1. Follow the procedures to sketch

(a) $y = \dfrac{x^2}{x^2+3}$

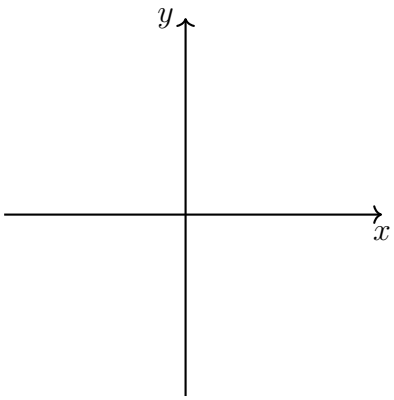
(b) $y = \dfrac{x}{(x-1)^2}$

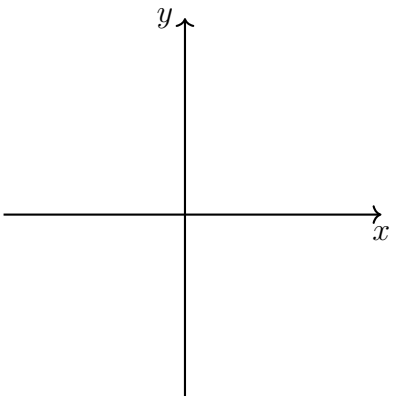




(c) $y = x\sqrt{1-x^2}$

(d) $y = x - \sqrt[3]{x}$





6 Reasoning and Analyzing

Each statement is either **always true, sometimes true,** or **never true**. When you decide on your answer, you must provide a **justification**.

If you choose **sometimes true**, you must show one scenario when it is true, and another, when it is false.

For **always true** and **never true**, you must provide a brief explanation why. Sometimes this can simply be a definition, or sometimes it may require some more mathematical reasoning.

Example 6.1. For each of the following statements, identify if the statement is always true, sometimes true, or never true. Provide a brief justification.

a) If $f(x)$ has a horizontal asymptote at $y = L$, then $\lim_{x\to\infty} f(x) = L$.

b) If $f(x)$ is a twice differentiable function and $f''(a) = 0$, then $x = a$ is a point of inflection.

c) If $f'(x) > 0$ for all x, then $f(x)$ is concave up everywhere.

d) If $\lim_{x\to\infty} f(x) = L$, then $f(x)$ has a horizontal asymptote at $y = L$.

e) If $\lim_{x\to a} f(x) = \infty$, then $f(x)$ has a vertical asymptote at $x = a$.

f) If $f'(a) = 0$ and $f''(a) = 0$, then by the second derivative test, $x = a$ is neither a maximum or a minimum.

Practice Exercises

1. For each of the following statements, identify if the statement is always true, sometimes true, or never true. Provide a brief justification.

   (a) If $f(x)$ has a horizontal asymptote at $y = L$, then $\lim_{x \to -\infty} f(x) = L$.

   (b) If $f(x)$ is differentiable everywhere in its domain, then $f(x)$ is twice differentiable everywhere in its domain.

   (c) If the concavity changes over $x = a$, then $f''(a) = 0$.

   (d) If $h(x) = \dfrac{f(x)}{g(x)}$ and $g(a) = 0$, then $h(x)$ has a vertical asymptote at $x = a$.

   (e) If f is concave down everywhere, then $f'(x) < 0$

   (f) If $x = a$ is a local maximum, then $f''(a) < 0$.

   (g) If $f'(a) = 0$ and $f''(a) > 0$, then $x = a$ is a local minimum.

   (h) If $\lim_{x \to \infty} f(x) = L$, then there is no horizontal asympte.

7 Answer Key

Lesson 1 - Exercises

1. Find the vertical asymptotes.

   (a) $x = -1$ (b) $x = -1, x = 1$ (c) $x = \frac{1}{3}$

   (d) $x = \frac{3}{2}, x = -\frac{3}{2}$ (e) $x = -3$ (f) $x = -1$

2. Find the horizontal asymptotes.

   (a) $y = 1$ (b) $y = -2$ (c) $y = 0$

   (d) $y = 1$ (e) $y = 1$ (f) $y = -\frac{2}{9}$

Lesson 2 - Exercises

1. Find the intervals on which the curve is concave upward or concave downward, and state the points of inflection.

   (a) CU on $(-\infty, \infty)$
   No inflection points.

   (b) CU on $(-\infty, 0) \cup (1, \infty)$
   CD on $(0, 1)$
   Inflection Points: $(0, -2)$ and $(1, -2)$

   (c) CU on $(1, \infty)$
   CD on $(-\infty, 1)$
   No inflection points.

   (d) CU on $(\infty, -\sqrt{6}) \cup (0, \sqrt{6})$
   CD on $(-\sqrt{6}, 0) \cup (\sqrt{6}, \infty)$
   Inflection Points: $\left(-\sqrt{6}, \frac{5}{6\sqrt{6}}\right), \left(\sqrt{6}, \frac{-5}{6\sqrt{6}}\right)$

2. For each of the following

   i. $y = x\sqrt{x^2 + 4}$

   (a) Increasing on $(-\infty, \infty)$
   (b) No local max/min
   (c) CU on $(0, \infty)$
   CD on $(-\infty, 0)$
   (d) Inflection point: $(0, 0)$

   ii. $y = 3x^{\frac{2}{3}} - 2x$

   (a) Increasing on $(0, 1)$
   Decreasing on $(-\infty, 0) \cup (1, \infty)$
   (b) Local Max: $(1, 1)$, Local Min: $(0, 0)$
   (c) CD on $(-\infty, 0) \cup (0, \infty)$
   (d) No inflection points.

Lesson 3 - Exercises

1. Find the intervals on which the curve is concave upward or concave downward, and state the points of inflection.

   (a) min : $(0, 1)$
   max : $(1, 2)$

   (b) min : $(\sqrt{3}, 1), (-\sqrt{3}, 1)$
   max : $(0, 10)$

   (c) min : $\left(-3, -\frac{1}{6}\right)$
   max : $\left(3, \frac{1}{6}\right)$

   (d) max : $(-8, 4)$
   $x = 0$ inconclusive by SDT
   min at $(0, 0)$ by FDT

Lesson 4 - Exercises

1. Find the equation of the slant asymptote.

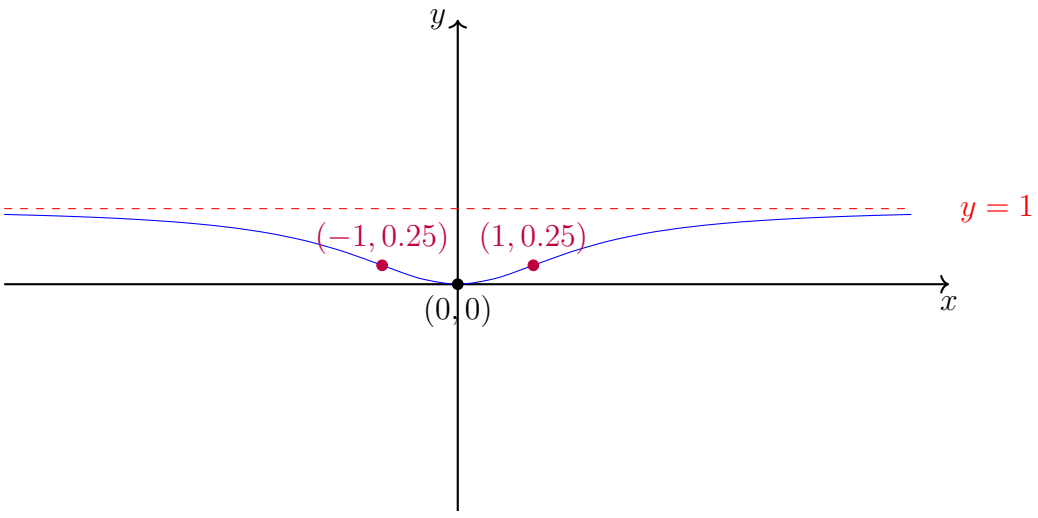
   (a) $y = -x + 2$ (b) $y = x$ (c) $y = 3x + 1$

   (d) $y = 2x - 1$ (e) $y = x + 4$ (f) $y = -x$

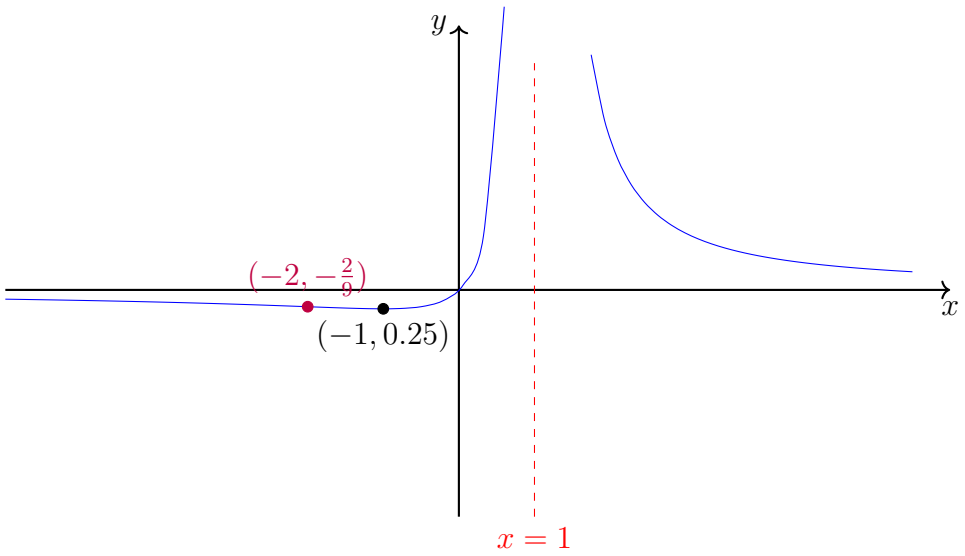
Lesson 5 - Exercises

1. Follow the procedure to sketch. **Inflection Points**

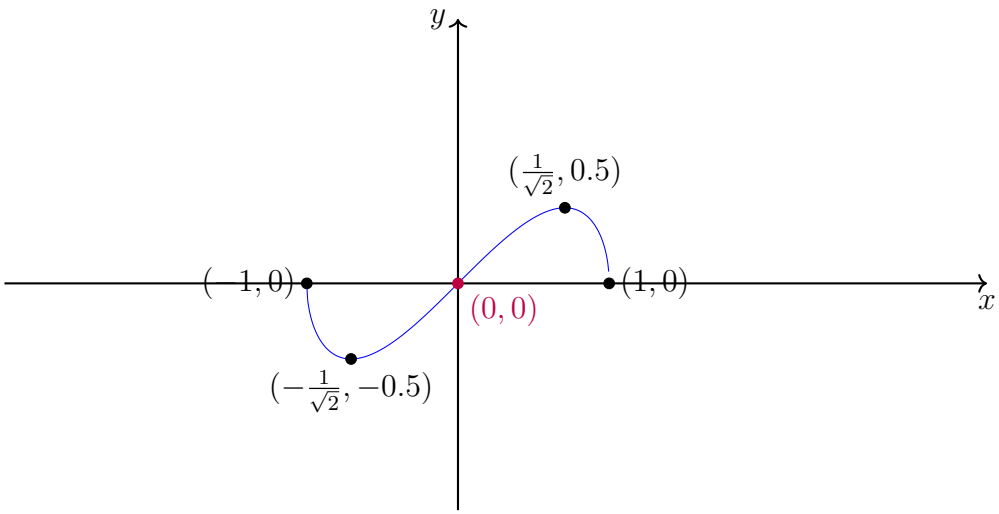
   (a)



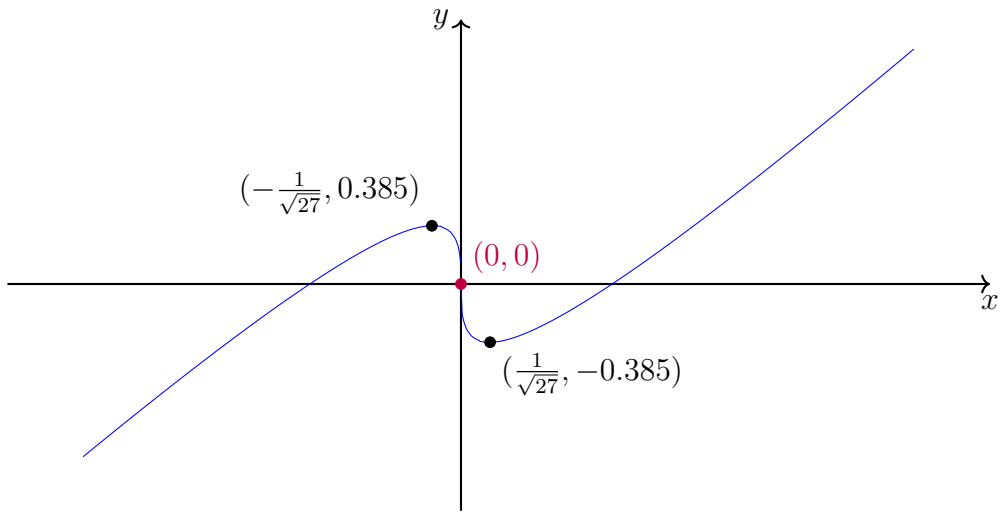
(b) ‘



(c)



(d)



Lesson 6 - Exercises

1. a) Sometimes true. $f(x)$ can have a horizontal asymptote if as $x \to \infty$, $f(x) \to L$. The left side does not need to have the same behaviour. For example, $y = \dfrac{1}{x}$ has the same horizontal asymptote on both sides. $y = (0.5)^x$ has a horizontal asymptote on the right end behaviour only.

   b) Sometimes true. $f(x) = x^{\frac{4}{3}}$ is differentiable everywhere, but not twice differentiable at $x = 0$. Any cubic polynomial is differentiable everywhere and also twice differentiable.

   c) Sometimes true. $x = a$ might be a vertical asympte. $y = \dfrac{1}{x}$ has left side concave down, and right side concave up, but the second derivative at $x = 0$ does not exist. $y = x^3$ would be an example where the concavity changes and the second derivative at $x = 0$ is zero.

   d) Sometimes true. $x = a$ might just be a point of discontinuity. $y = \dfrac{x^2 - 4}{x - 2}$ is a counterexample. $y = \dfrac{1}{x - 2}$ is an example.

   e) Sometimes true. Concave down can still have a part that is increasing. $y = -x^2$ is a counterexample. $y = -2^x$ is concave down everywhere and decreasing everywhere

   f) Sometimes true. If it was a corner, then f' and f'' does not exist, but it is still a maximum. If it was smooth, then true.

   g) Always true. Definition of Second Derivative Test.

   h) Never true. Definition of horizontal asymptote has been met.